JPMC AI-Assisted Connected Patient Flow System

Week-14 Installation Status Report · SDLC Build Phase (Wk 8–15)

Prepared for: Jinnah Postgraduate Medical Centre (JPMC), Karachi · **Prepared by:** Implementation Partner — Project Manager · **Report date:** 30 September 2026 · **Document v1.0**

WEEK
14 / 28

OVERALL VERDICT

GREEN

On track for Wk 24–28 go-live

83%	2 / 6	3 / 3	11 / 1 / 0
Build completeness	Sandbox connected	AI functions in eval	Open Sev-3 / Sev-2 / Sev-1

End of Week 14 of a planned 24–28-week SDLC implementation. The project sits in the back third of the Core Development window (Wk 8–15) with AI/ML Development (Wk 10–15) and Integration (Wk 12–18) running concurrently. All four clinical-safety principles remain enforced: AI organizes, predicts and flags only; clinicians retain diagnosis, triage and prescribing authority. No material slippage against the Week 1–3 baseline plan; one pharmacy sandbox credential request and one HIS patient-identity data-owner assignment need JPMC action this fortnight.

1. Schedule vs. Plan

The schedule below maps the nine SDLC phases from the approved Implementation Plan against current status at end of Week 14. Seven phases are complete or in progress; the Integration window opened in Week 12 and remains the critical concurrent path. No phase has slipped against its baseline; the Testing gate (Wk 16) is on track to be reached on schedule.

Phase	Weeks	Status	Notes at Week 14
Planning & discovery	1–3	COMPLETE	Charter, site survey, pilot scope, baseline, risk plan signed off.
Analysis	2–5	COMPLETE	User stories, data dictionary, KPI definitions, integration spec frozen.
Design	4–7	COMPLETE	UX prototypes, architecture, security/test designs reviewed by JPMC IT.
Core development	8–15	IN PROGRESS	Backend ~88%; portals/dashboards ~75–83%. Backlog on track for Wk 15 close.
AI/ML development	10–15	IN PROGRESS	Summary, wait-prediction, vitals flags in evaluation; bias review #2 due Wk 15.
Integration	12–18	IN PROGRESS	HIS sandbox live; billing in dev; pharmacy credentials pending JPMC IT.
Testing & security	16–19	UPCOMING	Traceability matrix 70% drafted; UAT scenario packs due Wk 15.
Beta & pilot deploy	20–23	UPCOMING	Internal alpha, controlled beta, pilot expansion per plan.
Training & go-live	24–28	UPCOMING	Hypercare 2–4 weeks; AMC starts at stabilization.

2. Core Build Progress

Backend API build is the most advanced track at approximately 88% complete. Patient identity/profile, registration, appointment/token, queue, notifications and document services are functionally complete; the audit-logging layer is fully wired against the RBAC matrix; encrypted document storage is in place using JPMC-approved key handling. Remaining backend work is concentrated on operational-report endpoints for the administration dashboard and on time-zone hardened scheduling for token issuance.

Frontend build tracks the backend by approximately one week, as expected. Patient portal/app: registration, token, queue status, prescription visibility and payment/discharge notices are at ~83%; follow-up reminders and report upload/preview are in code review. Doctor dashboard: secure sign-in, availability status, patient history, medication/allergy view and vitals display are at ~75%; the AI-organized summary pane is feature-flagged behind the evaluation endpoint. Nurse/triage station: quick identification, vital capture (manual and device) and triage priority routing are at ~80%; emergency escalation flow is in peer review. Administration dashboard is the least mature at ~70% — queue/workload view and user/role administration are live; operational reports and audit search remain.

RBAC enforcement, audit-log completeness and encrypted document storage are built and unit-tested; feature flags for all AI surfaces are wired so that disabling any AI function leaves the manual clinical workflow fully usable, per the safety principle. The next two prioritized build items for Week 15 are the queue-display interface for the OPD waiting area and the operational-report pack on the administration dashboard.

3. AI/ML and Integration Status

3.1 AI/ML development (Wk 10–15)

All three AI functions are in evaluation behind feature flags on the staging environment. The Patient Information Organizer (structured-summary workflow) is generating encounter summaries on test data; the clinician-validation workflow is live, and the human-review mechanism records every accept/edit/reject action against the source record. The Waiting-time Prediction model is trained on the baseline queue dataset captured in Weeks 2–5 and is in evaluation against the error targets agreed with the Clinical Lead. The Vitals Flagging rules engine is deployed with the configured thresholds for BP, pulse, temperature, SpO_2 and weight; flags route to a review queue rather than auto-escalating. The de-identified AI evaluation dataset is prepared; bias/safety review #1 is closed, review #2 is scheduled for Week 15. All AI outputs continue to display source information, timestamp and confidence/limitations, and the manual fallback path remains verified.

3.2 Integration layer (Wk 12–18)

The HIS sandbox adapter is connected; test patient, encounter and order messages are flowing in both directions and field mapping for the key FHIR resources is documented. The billing adapter is in development against the JPMC billing sandbox; charge-item and invoice endpoints are mapped, reconciliation logic is pending. Pharmacy adapter: sandbox credentials and API documentation were requested from JPMC IT on Week 13; this is the principal blocker on the integration track and is flagged in §5 of this report. Lab and radiology adapters are scoped and waiting on the pharmacy pattern to be finalized. The notification service (SMS / WhatsApp / push) is sandbox-tested end-to-end; production gateway selection is deferred to the change-control board.

4. Clinical Safety & Governance

The weekly Clinical Safety Review has been held every week since Week 4 and is on cadence; eleven reviews have been convened to date, each producing a minutes-and-actions note circulated to the Clinical Lead, Nursing/Triage Lead and JPMC IT/HIS Lead. The four AI-boundary rules from the Implementation Plan remain enforced in code and in review: no autonomous diagnosis, no autonomous prescription, no autonomous triage-severity assignment, and no overruling of clinician judgment. Every AI surface shows source record, timestamp, and confidence/limitations, and routes through an explicit clinician-accept/edit/reject workflow before any persistence or downstream propagation.

Triage workflow validation has been completed with the Clinical Lead and the Nursing/Triage Lead; escalation logic, documentation templates, digital prescription/test workflow and downtime forms are drafted and under review. The RBAC matrix, audit schema and privileged-access rules were reviewed and signed off by JPMC IT/Security in Week 12. The threat-model session for the application layer was completed in Week 13; one medium-severity finding on session-invalidation behaviour is open with a target close date of Week 16, ahead of the security testing window.

5. Risks, Blockers & JPMC Actions Needed

The table below updates the top six risks from the project Risk Register against current status. Items marked GREEN are mitigated and on track; AMBER require a specific JPMC action this fortnight; no risks are RED at Week 14.

#	Risk	RAG	JPMC action requested	Due
1	Integration delay / unavailable APIs (pharmacy, lab)	AMBER	Issue pharmacy sandbox credentials + API docs; assign JPMC IT integration owner.	Wk 15
2	Source data quality — HIS patient identity de-duplication	AMBER	Assign JPMC data owner; approve identity-matching rules; sign-off scoped migration.	Wk 16
3	Low staff adoption risk	GREEN	Confirm co-design workshop attendees (OPD, ER, nursing, billing) for Wk 16.	Wk 15
4	Network / power interruption at pilot sites	GREEN	Confirm site readiness review slot (Wk 17) and backup-power availability at OPD/ER.	Wk 17
5	Scope expansion — dental OPD request received	AMBER	Defer to change-control board; do not absorb into pilot scope.	Wk 16
6	Cybersecurity incident path — incident response readiness	GREEN	Confirm incident-response tabletop participants from JPMC IT/Security for Wk 18 session.	Wk 18

6. Next Two Weeks & Transition into Testing (Wk 15 → Wk 16–19)

Week 15 priorities are: (a) close the remaining core-dev backlog items — the queue-display interface for the OPD waiting area and the operational-report pack on the administration dashboard; (b) finalize the AI evaluation report and convene bias/safety review #2 with the Clinical Lead; (c) complete the pharmacy adapter sandbox connection once JPMC IT releases credentials; (d) finalize the test-design traceability matrix and the UAT scenario packs for named JPMC users. The fortnight closes with a biweekly steering-committee review where entry into the Testing window is formally approved.

Entry into the Testing window (Wk 16–19) is governed by the go/no-go gate defined in the approved Implementation Plan: zero open critical defects, security baseline passed, and AI evaluation acceptable to the Clinical Lead. On current trajectory all three gate criteria are projected to be met by end of Week 15. The Testing window will execute unit/API, system/integration, security (OWASP ASVS-style), performance (pilot concurrency, queue-update latency, dashboard response), usability/accessibility, AI evaluation and UAT in that order, with critical-defect closure required before any beta rollout activity begins in Week 20.

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